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Changes in proportion of bachelor's nurses associated with improvements in patient outcomes

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Abstract

This study uses data from two cross-sections in time (2006, 2016) to determine whether changes over time in hospital employment of bachelor's of science in nursing (BSN) nurses is associated with changes in patient outcomes. Data sources include nurse survey data, American Hospital Association Annual Survey data, and patient administrative claims data from state agencies in California, Florida, New Jersey, and Pennsylvania. The study sample included general surgical patients aged 18–99 years admitted to one of the 519 study hospitals. Multilevel logistic regression and truncated negative binomial models were used to estimate the cross-sectional and longitudinal effects of the proportion of hospital BSN nurses on patient outcomes (i.e., in-hospital mortality, 7- and 30-day readmissions, length of stay). Between 2006 and 2016, the average proportion of BSN nurses in hospitals increased from 41% to 56%. Patients in hospitals that increased their proportion of BSN nurses over time had significantly reduced odds of risk-adjusted mortality (odds ratio [OR]: 0.95, 95% confidence interval [CI]: 0.92–0.98), 7-day readmission (OR: 0.96, 95% CI: 0.94–0.99) and 30-day readmission (OR: 0.98, 95% CI: 0.95–1.00), and shorter lengths of stay (incident rate ratio [IRR]: 0.98, 95% CI: 0.97-0.99). Longitudinal findings of an association between increased proportions of BSN nurses and improvements in patient outcomes corroborate previous cross-sectional research, suggesting that a better educated nurse workforce may add value to hospitals and patients.

KEYWORDS

education, health service research, nursing, outcomes research

1 | INTRODUCTION

In 2010, the Institute of Medicine (IOM) (now the National Academy of Medicine) published *The Future of Nursing* report which put forth a set of recommendations for the United States. Among the recommendations was the ambitious goal of increasing the proportion of registered nurses (RNs) with a bachelor's of science in nursing (BSN) degree to 80% by 2020 (IOM, 2011). This recommendation based upon research evidence grew from a long and controversial history about the multiple educational pathways—diploma, associate, and bachelor's degree—to obtain a RN license. Beginning in 1965, the American Nurses Association called for the BSN as the required educational qualification for professional nurse licensure by 1985, a standard that was not met. In 2003, a landmark study demonstrated

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for the first time that patients were significantly less likely to die following surgery when cared for in a hospital with higher proportions of BSN-prepared nurses (Aiken et al., 2003). In the years that followed, other studies in the U.S. and abroad have further documented the link between more favorable patient outcomes and higher hospital proportions of BSN nurses (Aiken et al., 2011; Aiken et al., 2014; Audet et al., 2018; Cho et al., 2015; Estabrooks et al., 2005; Haegdorens et al., 2019; Harrison et al., 2019; Haskins & Pierson, 2015; Kutney-Lee et al., 2013; Van den Heede et al., 2009; White et al., 2018).

The bulk of prior evidence supporting a transition to a larger proportion of the nursing workforce holding a BSN has been conducted in the cross-section. An exception is a single longitudinal study of Pennsylvania hospitals using panel data of two cross-sections in time: 1999 and 2006. This study showed that while there was no change over time in the average proportion of BSN nurses in hospitals (32.5% in 1999; 32.7% in 2006), there was considerable variation across hospitals (Kutney-Lee et al., 2013). Furthermore, Kutney-Lee and colleagues described that between 1999 and 2006, roughly 45% of U.S. hospitals reduced their proportion of BSN nurses, approximately 35% increased their proportion, and about 20% had no change. This study was the first to contribute evidence of the association of changes over time in hospital proportions of BSN nurses and patient outcomes. In hospitals that increased their proportion of BSN nurses over time, patients had significantly reduced mortality and failure-to-rescue (i.e., death following a complication).

A more recent longitudinal study of 737 U.S. hospitals using panel data from 2006 to 2016 shows that on average hospital proportions of BSNs have increased over time (from 44.2% in 2006; to 60.1% in 2016) and nurses in hospitals that increased their proportions of BSN nurses over time were less likely to give poor quality of care ratings (Sloane et al., 2018). It is not yet clear whether the changes observed between 2006 and 2016 also translate into better clinical outcomes for patients. Moreover, there is insufficient evidence of the economic value of transitioning to a greater BSN composition of the nurse workforce. Research from Yakusheva et al. (2014) used a nurse-level analysis at a single U.S. hospital to suggest the possibility of an economic value of employing BSN nurses. Patients who received at least 80% of their hospital care from a BSN nurse had lower odds of readmission and shorter lengths of stay (Yakusheva et al., 2014).

The purpose of this study was to determine whether hospital increases in the proportion of BSN nurses over time were associated with objective outcomes including reduced mortality, reduced readmissions, and shorter lengths of stay. Together, this evidence serves to corroborate previous research of BSN nurses and patient outcomes conducted in the cross-section, by using a longitudinal study design to evaluate the relationship between changes in BSN workforce and changes in patient outcomes, and to build a value case for growing a workforce with greater numbers of BSN nurses consistent with IOM recommendations.

2 | METHODS

The data analyzed here are from three sources collected in two periods a decade apart. These sources included: (1) Survey data collected from a panel of 519 adult acute care hospitals at two points in time, in 2005-2006 and in 2015–2016, which provides information on nursing resources in the study hospitals including nurse education (i.e., RN4CAST). (2) Patient data from the same periods for general surgical patients discharged from these same hospitals, which provide information on mortality and lengths of stay for individual patients in all 519 hospitals and readmissions for patients in a subset of 371 hospitals, as well as patient characteristics, diagnostic related groups (DRGs), and comorbidities used to risk adjust them in each hospital in both periods. (3) The American Hospital Association (AHA) Annual Survey data for 2006 and 2016, which provide additional information on other hospital characteristics, including organizational and structural measures that may be associated with changes in the nurse educational composition measures and patient outcomes in the study hospitals. The data sources and their corresponding time periods are shown in Figure 1.

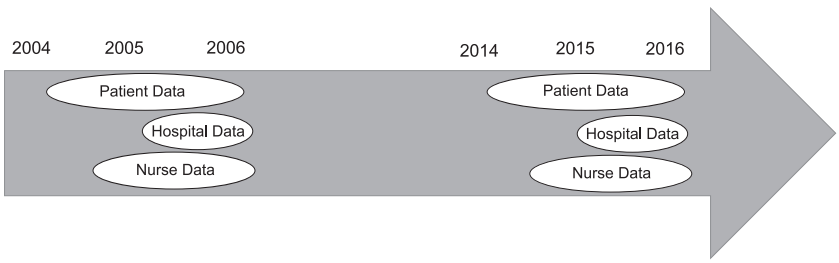

FIGURE 1 Study data sources. Patient data are obtained from state administrative claims data representing all inpatient hospital discharges within each study state (California, Florida, New Jersey, Pennsylvania). Nurse data are from the RN4CAST survey of nurses. Hospital data are from the American Hospital Association Annual Survey

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2.1 | Hospital sample, survey data, and measures

Traditionally, researchers interested in studying hospitals either cross-sectionally or longitudinally begin by constructing a sampling frame of all hospitals of interest and then randomly sampling some fraction of the hospitals in the sampling frame using a simple, clustered, or stratified approach. The problem with this traditional sampling strategy is that the participation of hospitals resides in the authority of hospital administrators. If hospital officials choose not to participate on the basis of the nursing features being studied—if, for example, administrators of hospitals with comparatively few BSN nurses are less likely to participate in a study centered on the educational credentials of nurses—then hospital level nonresponse would translate into a biased sample of hospitals.

Given our concern with nonresponse at the hospital level, we opted instead to directly survey a large sample of nurses using state licensure lists as sampling frames. We invited these nurses to complete a mailed survey which asked them whether they worked in a hospital and, for those who did, to choose the name of their hospital from a list of all hospitals in their state. We also asked them for information about their hospitals, including numerous items about themselves, including most importantly for current purposes their highest educational degree. Since funding constraints made it impossible to do this for all states in the U.S., we chose to focus on four large states—California, Florida, New Jersey and Pennsylvania—which are, like the U.S. as a whole, diverse with respect to urban and rural hospitals, and which include roughly 25% of Medicare beneficiary hospital discharges annually. Ultimately, we obtained information from nurse informants for virtually all (97%) of the 766 adult acute care hospitals in these states (Lasater et al., 2019).

In this study, we restrict the hospital sample to 519 hospitals with 10 or more nurse informants in both periods to render our estimate of the percentage of nurses with a BSN or higher degree more reliable. The average numbers of nurse informants per hospital were 67 nurses and 40 nurses in 2005-2006 and 2015–2016, respectively. We employed a double-sampling nonresponder survey approach to ensure nurses were unbiased informants with respect to their education. The nonresponder surveys undertaken in both periods after the main survey indicated that nurse informants were reasonably representative of all nurses in these hospitals in general, since we found very few significant differences between the responding nurses used as informants and nurses who did not respond to the main survey. Of relevance to this study, there were no significant differences in the education of nurses who responded in the main survey and those who participated in the nonresponder survey (Lasater et al., 2019).

Data from nurse informants are aggregated to provide measures of the nursing resources in each hospital in each period. Nurse education, our primary variable of interest, is analyzed as an organizational characteristic of the hospital—not the individual nurse—and measured as the percentage of RNs in each hospital that reported that their highest degree was a BSN or higher. Nurse staffing was measured by dividing the number of patients on each of the nurses' shifts by the number of nurses and averaging them in each hospital. The work environment was measured by the practice environment scale (PES), a National Quality Forum endorsed quality measure that has been widely used and extensively validated nationally and internationally (Aiken et al., 2008; Lake, 2002). The PES is a 31-item scale that asks nurse informants to indicate the degree to which various organizational features are present in their practice setting. We aggregate scores from 4 of the 5 subscales derived from these items to create a single summary measure of the practice environment for each hospital. The four subscales include (1) nurse participation in hospital affairs, (2) nursing foundations for quality care, (3) nurse manager ability, leadership, and support of nurses, and (4) collegial nurse-physician relations. The fifth subscale (staffing and resource adequacy), is excluded for current purposes because of its substantial correlation with our staffing measure. Published internal consistency coefficients (Cronbach's alphas) for these subscales range from 0.71 to 0.84. Intraclass correlation coefficients range from 0.86 to 0.96, well within the range of generally accepted values (Lake, 2002).

2.2 | Patient sample, data, and measures

The patient data used in this study are for adult surgical patients 18–99 years old discharged from the 519 study hospitals over two 3-year periods, 2004–2006 and 2014–2016, which were proximate to the hospital nurse surveys. The data were obtained from each of the four state agencies, including the Office of Statewide Health Planning and Development in California, the Agency for Health Care Administration in Florida, New Jersey Department of Health and Senior Services, and the Pennsylvania Health Care Cost Containment Council (PHC4). These agencies collect information from administrative claims data for all inpatient hospital discharges within each state. The patient-level data include hospital identifiers, patient demographics, admission information, payer, discharge status (alive/dead) and destination, and DRG assignment. In addition, International Classification of Diseases, Ninth Edition, Clinical Modification (ICD-9-CM) codes (and ICD-10-CM beginning in FY 2016) are recorded for both the principal diagnosis and principal surgical procedure, and up to 30 secondary diagnoses and procedures.

The discharge data provided the outcome measures for the study, which included 30-day in-hospital mortality, 7- and 30-day readmissions, and length of stay, as well as the patient characteristics used to risk adjust them (i.e., age, sex, transfer status, comorbidities, and surgical base-DRGs). In our analyses of mortality, we included all general surgical discharges, or roughly 1.64 million cases (about 830,000 cases in each period). For readmissions, about 932,000 of these cases were used, since we lacked sufficient information in one state (i.e., Florida) to code readmissions and we restricted our analyses to routine (or home) discharges. For length of stay we used 1.61 million cases, since we omitted about 54,000 (or 3%) of the cases that involved in-hospital deaths, admissions of transfer patients, and lengths of stay longer than 30 days.

2.3 | American hospital association (AHA) data

The AHA Annual Survey data for 2006 and 2016 provided information on the control variables in our analyses, related to study

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TABLE 1 Characteristics of the 519 Study Hospitals, 2006 and 2016

Hospital characteristic	2006 Mean	2006 *SD*	2016 Mean	2016 *SD*
Percentage BSN nurses	41%	12%	56%	15%
Nurse staffing (patients per nurse)	4.9	1.6	4.4	1.0
Nurse work environment score	2.7	0.2	2.8	0.2
	N	%	*N*	%
Teaching status				
Nonteaching	264	50.9	215	42.3
Minor teaching	214	41.2	241	47.4
Major teaching	41	7.9	52	10.2
Total	519	100.0	508	100.0
Technology status				
Not high technology	275	53.0	233	45.5
High technology	244	47.0	279	54.5
Total	519	100.0	512	100.0
Size (number of beds)				
100 or fewer	42	8.1	35	6.7
101–250	234	45.1	217	41.8
251 or more	243	46.2	267	51.5
Total	519	100.0	512	100.0

Abbreviation: BSN, bachelor's of science in nursing.

hospital size (i.e., number of beds), teaching status (hospitals with and without any postgraduate medical residents or fellows were dummy coded), and technology capabilities (which contrasted hospitals that did and did not perform open-heart surgeries or major organ transplants).

2.4 | Statistical analysis

We describe, using means and percentages, the nursing resources and other characteristics of the study hospitals in both periods. We show graphically the changes in hospital percentages of BSN nurses that took place over time and the percentages of hospitals that experienced increases and decreases in the percentage of BSN nurses. We show the outcomes and other characteristics of the patients discharged from these hospitals in the two periods, including characteristics needed to risk adjust them.

To evaluate the effects of the percentages of BSN nurses and changes in the percentages of BSN nurses on cross-sectional differences across hospitals and longitudinal changes within hospitals in 30-day mortality, 7-day readmissions, and 30 day readmissions, we used multi-level logit regression models with random effects, with patients at level 1 and hospitals at level 2. To evaluate the effects of the percentages of BSN nurses and changes in the percentages of BSN nurses on cross-sectional differences across hospitals and longitudinal changes within hospitals on length of stay—which is a count variable rather than a binary outcome—we used multi-level truncated negative binomial models. We treat the differences and changes in the percentages of BSN nurses as continuous measures. Both the logit models and negative binomial models allow us to estimate the associations between the longitudinal changes in percentages of BSN nurses and the changes in outcomes within hospitals, net of the cross-sectional associations of those variables between hospitals.

We show odds ratios (ORs), 95% confidence intervals (CIs), and *p* values associated with the cross-sectional and longitudinal effects of BSN nurses on the mortality and readmission outcomes and incident rate ratios (IRRs), 95% CIs, and *p* values for length of stay, first using an "unadjusted" model that estimated these effects simultaneously controlling for the change in the outcomes over time, and then using "adjusted" models which include controls for patient and hospital characteristics. Patient characteristics included age, sex, transfer status, and dummy variables for 29 comorbidities from the Elixhauser comorbidity index and for 28 base-DRGs for surgical patients. Hospital characteristics include size, teaching status, technology, and cross-sectional and longitudinal effects of the nurse work environment and nurse staffing.

3 | RESULTS

Table 1 shows that, overall, the average percentage of RNs with BSN degrees increased substantially across hospitals over the period, from 41% (in 2006) to 56% (in 2016). Nurse staffing, or the average number of patients cared for by nurses on any given shift, has improved somewhat, from 4.9 to 4.4 patients per nurse (inclusive of intensive care unit staffing), and the variability on average nurse staffing across hospitals, as indicated by the *SD*, has decreased markedly, from 1.6 to 1.0 patients per nurse. Nurse work environments changed little, if at all.

The change in the average percentage of BSN nurses across hospitals suggests improvements over time but provides little information on how the distribution of hospitals with respect to that percentage differed in the two periods and whether the change was similar or dissimilar across hospitals. Figure 2 shows this graphically. The left panel of Figure 2 reveals that the percentage of BSN nurses across hospitals was reasonably normally distributed in both periods, though the distribution shifted to the right; while it was centered around the modal categories of 30%–39% and 40%–49% in the first period, it was centered around the modal categories of 50%–59% and 60%–69% in the second. Even in the second period, however, less than 5% of the hospitals reached the IOM target of 80% or more BSN nurses. The right panel of Figure 2 shows that 12% of the hospitals decreased in their percentage of BSN nurses, and half of them decreased by 5% or more. The remaining hospitals—roughly 90% of the hospital sample - increased in their percentage of BSN nurses, and while the increase was small (less than 5%) for about 12% of the hospitals, more than two-thirds of the hospitals exhibited an increase in the percentage of BSN nurses of 10% or more, and more than one-third of the hospitals increased by 20% or more.

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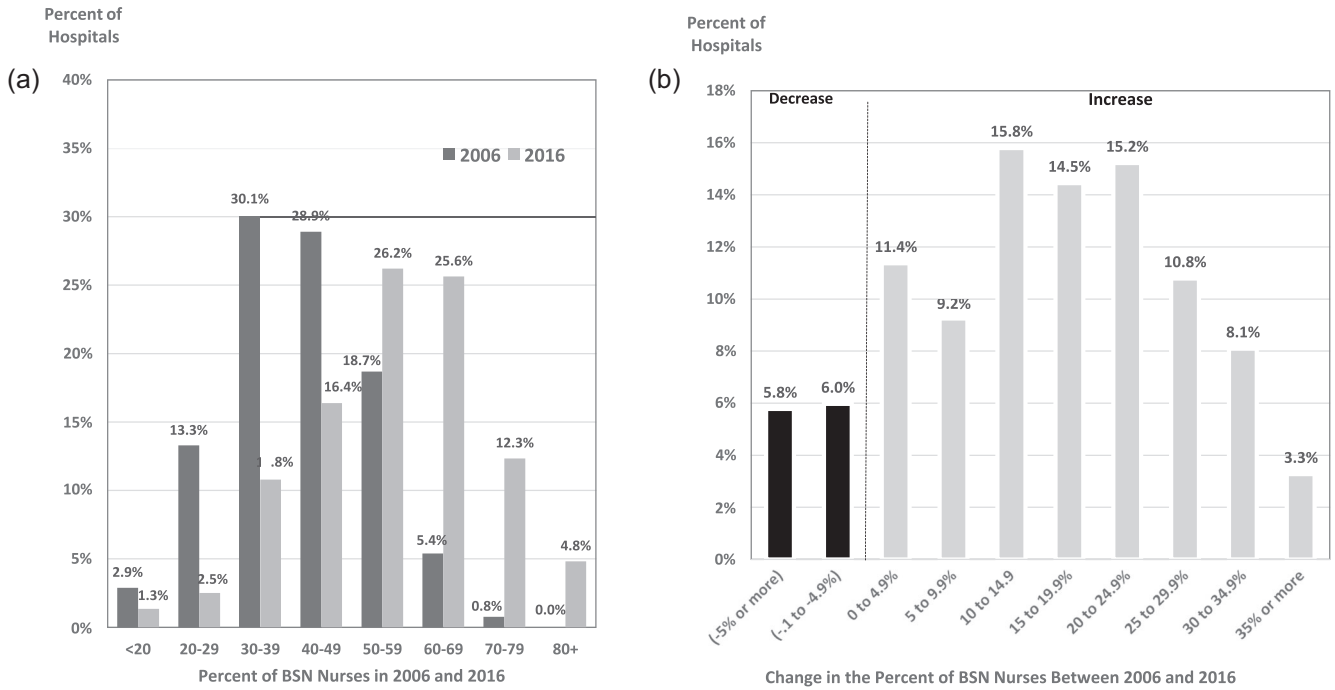

FIGURE 2 Percentage of hospitals with (A) varying percentages of BSN nurses in 2006 and 2016, and (B) varying decreases and increases in the percentage of BSNs over the period. BSN, bachelor's of science in nursing

With regard to patient outcomes, Table 2 shows that mortality among general surgical cases declined, from 1.4% to 0.9% over the period. Length of stay also diminished, to a lesser extent, from an average 5.1 days to 4.7 days, while both 7-day and 30-day readmissions remained essentially the same over the period, at about 3% and 7%, respectively. These favorable changes in mortality and the stability in readmissions occurred in spite of virtually all of the comorbidities increasing.

Table 3 provides odds ratios and incident rate ratios from unadjusted and adjusted models which show the cross-sectional differences in patient outcomes between hospitals that result from differences in hospitals in their percentage of BSN nurses, and the longitudinal changes (or changes over time) in outcomes that result from changes within hospitals in their percentage of BSN nurses. In all models, the percentage of BSN nurses was rescaled to make a single unit change equal to a change of 10 percentage points. The results from the unadjusted models, which estimate the cross-sectional and longitudinal effects simultaneously but control for no other effects except the general period effect (or the overall change in the different outcomes over time), are shown largely for comparative purposes and not amenable to any reasonable interpretation since they assume no other factors are at play, potentially confounding the associations. The adjusted models, by contrast, control not only for the overall change in each outcome over time but also for hospital and patient characteristics. The hospital-level characteristics include changes in hospital nurse staffing and nurse work environments, as well as hospital size, teaching status, technology, and location, while the patient characteristics include age, sex, transfer status, and dummy codes for 29 comorbidities and 28 surgical base-DRGs.

When these potential confounds are taken into account, we find a significant cross-sectional effect of the percentage of BSN nurses on mortality, a marginally significant cross-sectional effect ($p = 0.09$) of the percentage of BSN nurses on 30 day readmissions, but no significant cross-sectional differences in 7-day readmissions or length of stay associated with the percentage of BSN nurses. For all four outcomes, however, we find evidence of a significant change associated with changes within hospitals in the percentage of BSN nurses. While the size of the coefficients may seem fairly small (i.e., close to 1.0), they are quite sizable when we recall that a unit increase in the percentage of BSN nurses is equivalent to a 10 percentage point increase, and that a sizable number of hospitals witnessed increases in their proportions of BSN nurses of 20% or 30% or more. With respect to mortality, the odds ratio of 0.95 associated with a 10 percentage point increase in BSN nurses implies that the decrease in mortality for patients in hospitals which had a 20% increase in BSN nurses would be greater than for those hospitals that did not change at all by a factor of $0.95^2 = 0.90$ or by 10%. Even the smallest odds ratio we find, of 0.98 for the change in 30-day readmissions, would imply that the decrease in readmissions for patients in hospitals which had a 20% increase in BSN nurses would be greater than for those in hospitals that did not change at all by a factor of $0.98^2 = 0.96$ or by 4%. We found no evidence that hospitals' baseline proportion of BSN nurses was related to the estimated longitudinal effects on patient outcomes.

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TABLE 2 Outcomes and selected characteristics of general surgical patients in the Study Hospitals, 2006 and 2016

General surgical patient outcomes	2006 Numbers	2006 Percent	2016 Numbers	2016 Percent
Surgical cases/deaths	812,567/11,309	1.4%	825,703/7,627	0.9%
Surgical cases/7-day readmissions	476,974/15,087	3.2%	439,169/14,004	3.2%
Surgical cases/30 day readmissions	476,974/32,880	6.9%	30,960	7.0%
	Number of cases	**Mean (*SD*)**	**Number of cases**	**Mean (*SD*)**
Surgical cases—length of stay	785,533	5.1 (4.9)	798,536	4.7 (4.5)
General surgical patient characteristics				
Age	812,567	54.7 (18.4)	825,703	55.5 (17.9)
	Number of cases	**Percent (%)**	**Number of cases**	**Percent (%)**
Sex				
Female	491,667	60.5	484,623	58.7
Male	320,886	39.5	341,072	41.3
Total	812,553	100.0	825,695	100.0
Common comorbidities				
Hypertension	300,968	37.0	393,497	47.7
Diabetes without chronic complications	105,973	13.0	126,678	15.3
Chronic obstructive pulmonary disease	102,613	12.6	120,434	14.6
Anemia deficiency	67,202	8.3	111,556	13.5
Fluid and electrolyte disorders	99,773	12.3	156,442	18.9
Hypothyroidism	55,014	6.8	86,710	10.5
Obesity	56,461	6.9	127,248	15.4
Depression	46,132	5.7	76,993	9.3
Metastatic solid tumor	46,007	5.7	41,573	5.0
Renal disease	21,978	2.7	59,334	7.2
Diabetes with chronic complications	15,350	1.9	45,985	5.6

Notes: The percentage of deaths are for all general surgical cases. Readmissions are based on cases that exclude cases from Florida and that did not involve routine (home) discharges. Cases used to calculate length of stay exclude cases involving in-hospital deaths, admissions of transfer patients, and lengths of stay longer than 30 days. Comorbidities shown are those that involved at least 5% of the patients in either of the two periods, and do not sum to 100% due to patients with no comorbidities and multi-comorbidities.

4 | DISCUSSION

In this study we found that in a large sample of representative U.S. acute care hospitals, the proportion of BSN nurses, on average, increased between 2006 and 2016. This change, from an average of 41% BSN nurses (in 2006) to 56% BSN nurses (in 2016) represents the largest change in nursing resources over the decade, as staffing and work environment changes have been more modest. Despite these improvements, there remains large variability in hospital proportions of BSN nurses in 2016 ranging from hospitals with less than 20% BSN to those with more than 80%. Less than 5% of hospitals reached the 80% or greater levels of BSNs recommended by the IOM, and 12% of hospitals declined during the decade in proportion of BSNs. Thus, while there is progress in moving toward a greater proportion of BSNs in hospitals, there remains considerable room for improvement. Consistent with previous research (Aiken et al., 2011; Aiken et al., 2014; Aiken et al., 2003), we found in our current cross-sectional analysis that higher proportions of BSN nurses in hospitals were associated with better patient outcomes such as lower odds of mortality.

Our study adds new findings to what is currently known about the associations of BSN nurses and patient outcomes. In evaluating the longitudinal relationship, we found that patients in hospitals improving their BSN proportions over time had reduced odds of mortality, reduced odds of readmission (both 7 and 30-day) and shorter lengths of stay, even after adjusting for potentially confounding factors related to hospital and patient characteristics.

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TABLE 3 Coefficients from unadjusted and adjusted models indicating the cross-sectional differences across hospitals and longitudinal changes within hospitals in mortality, readmissions, and length of stay associated with the percentage of BSN nurses

	Unadjusted			Adjusted		
Patient outcome	**Odds ratio**	**95% CI**	***p***	**Odds ratio**	**95% CI**	***p***
Mortality						
Cross-sectional	0.899	0.876–0.922	<0.0010	0.929	0.902–0.956	<0.0010
Longitudinal	0.976	0.943–1.009	0.1587	0.948	0.916–0.981	0.0023
30-Day readmission						
Cross-sectional	1.014	0.995–1.034	0.151	0.986	0.969–1.002	0.0908
Longitudinal	0.983	0.960–1.007	0.1676	0.978	0.953–0.999	0.0384
7-Day readmission						
Cross-sectional	1.004	0.983–1.026	0.6793	0.988	0.969–1.008	0.2302
Longitudinal	0.968	0.939–0.997	0.0306	0.963	0.936–0.991	0.0112
	IRR		***p***	**IRR**		***p***
Length of stay						
Cross-sectional	0.960	0.944–0.977	<0.0010	1.000	0.988–1.013	0.9872
Longitudinal	0.989	0.972–1.006	0.2041	0.983	0.972–0.995	0.0038

Note: In all models the percentage of BSN nurses was rescaled to make a single unit difference or change equal to a difference or change of 10 percentage points. Unadjusted coefficients estimating the associations between the percentage of BSN nurses and the outcomes are from models which control only for period (to allow for changes in the outcomes over time). Adjusted coefficients are from multi-level models which also control for hospital and patient characteristics. The hospital-level characteristics include differences (and changes) in hospital nurse staffing and nurse work environments, as well as hospital size, teaching status, technology, and state. Patient characteristics include age, sex, transfer status, and dummy codes for 29 comorbidities and 28 surgical base diagnostic related groups.

Abbreviations: BSN, bachelor's of science in nursing; CI, confidence interval; IRR, incident rate ratios.

Prior hospital-level research has only found associations between nurse education and patient mortality. Our readmissions and length of stay findings suggest there may be a value case for hospitals to employ larger proportions of BSN nurses as fewer readmissions and shorter lengths of stay translate into cost savings for hospitals and payers. In addition to the favorable patient outcomes associated with BSN nurses, hospitals have strong incentive to preferentially hire BSN nurses, since the salary differential between BSN and associate degree nurses is modest, if any (Buerhaus et al., 2017; Chu et al., 2017). The public costs of BSN education are also relatively minimal as pre-licensure nurses typically pay for their own tuition and clinical training. Future research that includes cost-benefit analyses would strengthen the evidence that increasing hospital BSN compositions results in cost savings.

Our findings have a number of policy and practice implications and add to a body of knowledge driving both government and organizational policy initiatives that encourage growth in the number and employment of BSN nurses. For example, New York's "BSN in 10" regulation requires every RN to obtain a BSN within 10 years of initial RN licensure (Newland, 2018). While New York serves as a legislative example of efforts to reach the 80% IOM goal, preferential hiring practices are the main driver of increases in the proportion of BSNs in the U.S. (AACN, 2019). Nursing school leadership estimates that over 80% of employers show strong preference for hiring BSN nurses while 43% of hospital and other healthcare setting employers require all new nurse hires to hold BSNs (AACN, 2019). Since a BSN degree is a requirement for entry into graduate-level education, expanding baccalaureate completion programs is a possible pathway for increasing the numbers of BSN nurses in the U.S., as well as increasing the numbers of nurses with graduate degrees and nursing faculty. North Carolina's state policy standardizes minimum pre-requisites for RN to BSN programs with the goal of reducing barriers for associate degree nurses to obtain a BSN (North Carolina Board of Governors, 2015).

Nursing education trends align with these organizational and government policies as there were as many as 140,000 BSN graduates in 2018, 47.5% of which were already licensed associate degree nurses returning to obtain their BSN (Campaign for Action, 2019). This compares to 2010 when there were only 81,000 BSN graduates of which 31% were nurses with associate degrees obtaining a BSN (Campaign for Action, 2019).

The findings in our study also corroborate previous cross-sectional findings, which suggest that the large evidence base of cross-sectional research is consistent with longitudinal studies of panel data and changes in hospital nursing workforce over time. In particular, the effect size of a 10% increase in the proportion of BSN nurses on mortality which we observed in our longitudinal analysis (i.e., OR: 0.95; 95% CI: 0.92–0.98) is the same magnitude as that found in the seminal cross-sectional study of 168 Pennsylvania hospitals using data from 1999 (Aiken et al., 2003).

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Twenty years after research first documented that better patient outcomes were associated with higher proportions of BSN qualified nurses, we continue to find wide variability in the proportion of BSNs by hospitals. One in ten hospitals have decreased their proportions of BSNs, which suggests that hospital leadership in some institutions is lagging in evidence-based workforce decision-making. While disparities in the geographic distribution of BSNs may create supply issues for some hospitals to increase their proportions of BSN nurses—particularly those in rural areas (Odahowski et al., 2020; Smith et al., 2019), all hospitals can undertake strategies to increase their ability to attract BSN nurses, such as obtaining Magnet recognition (Kutney-Lee et al., 2015; McHugh et al., 2013) or creating their own supply by partnering with educational institutions like community colleges to develop BSN programs (Cheshire et al., 2017; Hawkins et al., 2018). Countries including Canada, Australia, New Zealand, Norway, Spain, and Philippines, all require a BSN as the standard entry to practice as an RN, yet the U.S. continues to lag in the development of a highly educated nurse workforce, with approximately 64% of nurses holding a BSN (HRSA, 2021).

4.1 | Limitations

Our longitudinal study design using panel data from two large representative groups of hospitals with repeated cross-sections of patients and nurses at two points in time advances current understanding of potential causal relationships between hospital nurse education composition and patient outcomes. As with all cross-sectional studies, we cannot rule out the possibility that unobserved and unmeasured factors other than changes in nurses' educational qualifications explain the results. Despite these limitations, our study has a number of strengths, including the evaluation of a large sample of hospitals and the patients within them from four large, geographically and demographically representative U.S. states. No other data exist to estimate hospital-level proportions of BSN nurses in such a large sample at multiple points in time. The adult general surgical patients in our analytic sample represent only 5.5% of the total hospital admissions in these study hospitals; and so, the true impact of nurse education on patient outcomes may be substantially underestimated here, since many patients potentially affected by better nurse education have not been included in the study.

5 | CONCLUSION

In this study, new evidence from panel data created a stronger case for the causal relationship between the proportion of BSN nurses in hospitals and lower odds of patient mortality, readmission and shorter lengths of stay. This evidence supports the ongoing pursuit of achieving the IOM recommendation of 80% BSN nurses, and should give confidence to hospital leaders to continue to increase employment of BSNs. We also document that while progress has been made towards this IOM goal, there is still much room for improvement.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests. The content is solely the authors' responsibility. This paper is our original, unpublished work and it has not been submitted to any other journal for review.

DATA AVAILABILITY STATEMENT

Data are available upon reasonable request. Data may be obtained from a third party and are not publicly available. US Nurse survey data: These data were collected under the NINR grant.

DISCLOSURE

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